Jessie Coleman

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EDUCATION

Emory University — Atlanta, GA
B.S. in Computer Science | May 2024
Relevant Courses: Data Structures/Algorithms; Linear Algebra; Calculus I/II/III; C Fundamentals

EXPERIENCE

Savoye NA — Associate Software Engineer, Philadelphia, PA | Jun 2025 – Present
- Built C++/OpenCV computer-vision tooling from scratch, implementing image-processing logic, model-output parsing, regression-based post-processing, confidence evaluation, runtime debugging, and performance-conscious detection of empty totes with 98% accuracy across repeated testing.
- Built a real-time 3D monitoring platform from scratch using C#, Unity WebGL, React, and WebSockets, replacing a legacy 2D interface and visualizing 10,000+ connected assets in the browser.
- Implemented Unity runtime systems for camera navigation, object selection, device search/filtering, status-driven visual changes, popups, scene hierarchy mapping, and live object-state visualization.
- Optimized runtime memory and performance by profiling large Unity scenes, reducing unnecessary update overhead, tuning object behavior, limiting expensive physics/runtime operations, and improving responsive WebGL rendering.

CWT Industries — Programmer, Atlanta, GA | Jan 2025 – Jun 2025
- Developed and debugged C#/.NET software for Caterpillar drilling systems, supporting PLC-integrated configuration, diagnostics, controls, and real-time monitoring.

Movement Labs — Junior Software Engineer, Oakland, CA | Mar 2024 – Jan 2025
- Built production software using React, Node.js, TypeScript, SQL, Docker, and Kubernetes, contributing to frontend features, backend services, debugging, and deployment support.

Publicis Sapient — Full-Stack Software Engineer Intern, Atlanta, GA | Jun 2023 – Aug 2023
- Developed React, Node.js, Java/Spring Boot, and AI-assisted application features for a client-facing recommendation platform; received Intern of the Month.

PROJECTS

Rolleroidz — Online Multiplayer 3D Video Game | Unity, C#, WebGL, JavaScript
- Built an online multiplayer 3D game that runs in the browser through Unity WebGL, with custom C# systems for player movement, physics simulation, camera behavior, animation handling, input handling, UI, player state management, and real-time match flow.
- Developed a custom cel-shaded rendering system, tuning stylized lighting, material behavior, visual readability, and real-time browser rendering performance for WebGL deployment.
- Built and debugged engine-adjacent systems including a custom physics solver, animation graph behavior, collision handling, movement feel, camera control, and multiplayer state synchronization.
- Investigated runtime failures through crash dump analysis, disassembly-level debugging, WebGL build troubleshooting, browser performance profiling, and memory/performance optimization.

Computer-Vision Video Game Automation — Python, OpenCV, Raspberry Pi
- Built a Raspberry Pi-based automation robot that plays a video game automatically by capturing live gameplay frames, detecting game-state cues with OpenCV, evaluating confidence, and converting visual decisions into real-time controller inputs through external hardware I/O.

A* Pathfinding Visualization — C/C++, Arduino
- Built an Arduino-based visualization of the A* pathfinding algorithm in C/C++, implementing heuristic search, node evaluation, path reconstruction, memory-conscious data handling, and step-by-step debugging on constrained hardware.

SKILLS

Languages: C++, C, C#, Java, Python, TypeScript, JavaScript, SQL
Game/3D: Unity, WebGL, 3D Math, Linear Algebra, Custom Rendering, Physics Solvers, Animation Graphs, Runtime Optimization
Engineering: Debugging, Algorithms, Data Structures, Object-Oriented Programming, Git, Memory/Performance Optimization, Crash Dump Analysis, Disassembly